10.2.2.5. Functions of a Multimeter

It can measure the following electrical and electronic quantities.

1. DC Voltage
2. DC Current
3. AC Voltage
4. AC Current (available in few meters only)
5. Resistance
6. Continuity Test
7. Checking the Transistor (NPN or PNP) (available in few meters only)
8. Collector Emitter Leakage Current (I_{CEO}) of a transistor (when base kept open)
9. Capacitance up to 20 µF (available in few meters only)
10. Frequency up to 20 kHz (available in few meters only)



Figure 10.9 Sensitivity of an analog multimeter

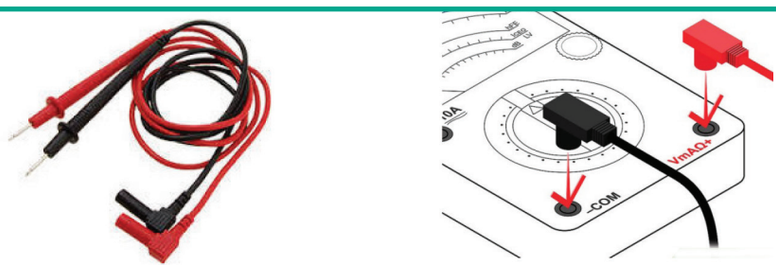
10.2.2.6. Sensitivity of an Analog Multimeter

A good quality analog multimeter offers an input impedance of 20 kΩ/V DC and 9 kΩ/V AC. This is known as sensitivity of an analog multimeter. It is printed on the bottom of the dial as shown in Figure 10.9. It means, to measure one volt DC, the meter offers 20 kΩ of impedance and to measure one volt AC, the meter offers 9 kΩ of impedance. Hence, the meter draws minimum current. Because of this, sensitive components such as transistors, IC etc., are not affected while testing with multimeter.

But low sensitivity meters (having an impedance below 5 kΩ/V AC and DC) draw high current from the circuit to be measured, which results in damage of transistors or ICs in circuit to be tested.

10.2.2.7. Precautions to be taken while handling an Analog Multimeter

A multimeter has one **positive probe** (Red colour) and a negative probe (black colour) to measure parameters as shown in Figure 10.10.



(i) Probes (ii) connecting probes with correct polarities

Figure 10.10 Connecting colour probes to correct polarities

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